

Protein Folding Is a Geometry Problem: Water Structure and Ion Tension Define the Folding Landscape

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Abstract

Protein folding has been treated primarily as a thermodynamic and sequence-driven chemical problem. This paper argues that framing is incomplete. The dominant physical forces shaping the folding landscape are not the intrinsic chemical properties of amino acids acting in isolation, but the geometry of water surrounding the chain and the tensional state set by dissolved ions. Water does not simply solvate; it organizes into structured geometries around partial charges and nonpolar surfaces that create spatially defined pressure gradients. Ions do not simply screen charge; they tune the tension of the local water geometry, setting rules about where and how fast segments can collapse. Together, these two factors — water geometry and ion tension — define a set of permissible folding windows: narrow spatial and temporal zones where the chain can sample and commit to a stable configuration. Misfolding is geometry drift. Aggregation is geometry failure at scale. Chaperones are geometry reset mechanisms. This paper develops that full argument from first principles, walks through propagation mechanics, explains wake-flow effects during folding, and shows how this geometry-first view makes the existing experimental data more coherent than the chemistry-first view does.

1. Folding Is Geometry, Not Chemistry

Biochemists have known for sixty years that sequence determines structure. The Anfinsen experiment settled that in 1961: denature ribonuclease A, strip away its structure, let it refold in buffer, and it comes back to the same native conformation. The information for folding is in the sequence. That result has not been seriously challenged since, and it should not be. But knowing what determines the outcome and knowing the physical mechanism by which that outcome is achieved are two entirely different things. Anfinsen told us the answer is in the sequence. He did not tell us what reads the sequence and executes the fold. That executor is physical geometry — specifically, the geometry of water surrounding the chain and the tensional state of that geometry set by dissolved ions.

The chemistry-first view of folding runs like this: amino acids have intrinsic chemical properties — hydrophobicity, charge, hydrogen bond donor and acceptor capacity — and those properties drive the chain toward the native structure by minimizing free energy. Nonpolar residues bury to avoid water. Polar residues form hydrogen bonds with each other and with water. Charged residues seek countercharges. These are all real forces. They are not being disputed here. The problem is that this framing describes the endpoints and the energetics without describing the physical mechanism that moves the chain from the unfolded state to the folded one. It is a thermodynamic accounting, not a mechanistic explanation.

Every amino acid residue along a polypeptide chain is in contact with water. There are no exceptions during the early stages of folding — even residues that will eventually be buried are, at the outset, touching water or within one or two molecular diameters of it. This means every residue is participating in a water geometry event before it participates in any intramolecular chemistry event. The first thing that happens to a residue as it exits the ribosome or enters solution is not that it finds a hydrogen bond partner or buries itself in a hydrophobic core. The first thing that happens is that water organizes around it. That

organization is geometrically specific, physically real, and mechanistically prior to every intramolecular contact that follows.

The core claim of this paper is this: the folding trajectory is written in the geometry of the water cage before the chain moves. The chain follows geometric gradients — spatial patterns of pressure, order, and tension — that are already present in the surrounding medium. Chemistry provides the specificity of individual contacts once the chain has navigated to the right position. But geometry provides the landscape the chain navigates to reach those contacts. The sequence determines structure because the sequence determines the water geometry around the chain, and the water geometry defines the folding landscape. That is the physical mechanism Anfinsen's experiment implied and did not specify.

This view does not contradict thermodynamics. Free energy is still minimized. The native state is still the lowest free energy configuration available to the chain under physiological conditions. What this view adds is the physical mechanism underneath the thermodynamic description — the actual machinery by which the chain finds that minimum without searching blindly through conformational space, and the actual reason that machinery can fail in characteristic, predictable ways. Thermodynamics tells you where the chain ends up. Geometry tells you how it gets there and what happens when it goes wrong.

2. How Water Geometry Builds the Folding Landscape

Water is not a passive solvent. This point is stated frequently in biophysics and then immediately set aside when folding models are constructed. It should not be set aside. Water is a three-dimensional dynamic lattice of hydrogen bonds, and the geometry of that lattice changes based on what it is touching. The lattice is not uniform. It is spatially inhomogeneous in any solution containing a solute, and a protein chain in solution is an enormously complex solute presenting a spatially varying surface of polar groups, nonpolar

patches, charges, partial charges, and flexible backbone segments. Each of these surface features generates a characteristic local water geometry.

Around polar and charged residues, water dipoles align into relatively ordered, high-coordination structures. The water molecules in the first and second hydration shells around a charged group are not tumbling freely — they are oriented, constrained by the electrostatic field of the charge, and under a different local pressure than bulk water. This is well established experimentally through neutron scattering, X-ray scattering, and femtosecond infrared spectroscopy. The water around a glutamate residue is not the same as bulk water in its geometry, its dynamics, or its mechanical properties. It is stiffer. It is more ordered. It resists deformation more strongly.

Around nonpolar surfaces, the situation is different and in some ways more mechanically consequential. Water cannot form hydrogen bonds with nonpolar surfaces, so it is faced with a geometric problem: it must maintain its hydrogen bond network in the vicinity of a surface that cannot participate in that network. The solution water adopts is to reorganize — to curve the network around the nonpolar surface in clathrate-like structures, or in extended proteins, to simply exclude itself from regions where it cannot hydrogen bond effectively. This exclusion is not passive. It is entropically driven and mechanically active. The excluded water creates a pressure gradient at the nonpolar surface — a tendency to push the nonpolar surface away from the bulk, which is precisely the hydrophobic force as expressed geometrically rather than chemically.

The key concept here is the geometric gradient. When a protein chain is in solution, different segments of the chain experience different local pressures because they are surrounded by water with different geometric properties. A segment flanked by charged residues sits in high-order, mechanically stiff water. A segment flanked by nonpolar residues sits in excluded-volume, high-pressure water. A segment at the boundary between these two types experiences a sharp geometric gradient — a spatial transition in local pressure and water order. These gradients are not random. They map onto the sequence because each residue type generates a characteristic water geometry, and the sequence is a

specific linear arrangement of residue types that produces a specific three-dimensional pattern of geometric gradients when placed in water.

The geometric folding landscape is this pattern of gradients. Before any intramolecular contact forms, before the chain has moved a single bond rotation away from the fully extended conformation, the surrounding water geometry has already defined a bias field — a three-dimensional map of where segments will experience pressure to move, in which directions, and on what timescale. This is not metaphor. It is the physical consequence of the fact that water has structure, that structure depends on what it is touching, and that structure has mechanical properties. The landscape is physically present in the water from the moment the chain enters solution.

Secondary structure elements are geometry-stabilized first, and their chemistry follows. An alpha helix is conventionally described as a structure stabilized by hydrogen bonds between backbone carbonyl oxygens and amide nitrogens four residues apart. That is accurate. But the question is why those hydrogen bonds form in that particular geometry rather than some other. The answer is that helical geometry concentrates the hydrogen bonding in a way that expels water from the helix interior. The geometry of water exclusion is what makes the helix favorable — the interior of a helix is a low-entropy, structured environment that costs water less to exclude than to accommodate. The hydrogen bonds form because the geometry makes them the lowest-energy option, not the other way around. Similarly, beta strands form antiparallel sheets because that geometry minimizes the solvent-exposed surface area of backbone amide groups more efficiently than any other extended configuration. The water geometry prefers the sheet. The hydrogen bonds are the chemical consequence of that geometric preference. They are real. They are stable. But they are downstream of the geometry.

3. Ion Tension Sets Local Rules

Dissolved ions are not spectators in the folding process. Every ion in solution carries its own hydration shell — a structured layer of water molecules whose geometry is determined by the ion's charge, size, and charge density. Sodium, potassium, magnesium, calcium, sulfate, chloride — each one creates a distinct local water geometry. When these ions sit near a segment of an unfolded polypeptide chain, they do not simply screen electrostatic interactions. They restructure the water geometry around that segment. And that restructuring changes the mechanical properties of the medium the chain is folding through.

Ion tension, as defined here, is the net mechanical stress in the water lattice around a chain segment, set by the concentration, species, and spatial distribution of ions near that segment. High ion tension means the water geometry around a segment is under stronger organizational stress — the hydrogen bond network is more constrained, more ordered, less able to rearrange rapidly. Low ion tension means looser, more permissive geometry — the water can rearrange more freely, the network is less constrained, and the chain has more conformational options available to it in that local region.

The Hofmeister series is the clearest experimental evidence that ion identity matters for protein stability in ways that go beyond electrostatic screening. Ions at the kosmotropic end of the series — sulfate, phosphate, fluoride — stabilize folded proteins and promote precipitation of unfolded ones. Ions at the chaotropic end — guanidinium, perchlorate, thiocyanate — destabilize folded proteins and promote unfolding and denaturation. The conventional explanation involves direct ion binding to the protein surface, changes in surface tension, and preferential hydration. All of those effects are real. But the geometry-first view provides a unifying mechanism: kosmotropic ions enhance water structure globally and locally. They tighten the geometry of the water lattice. They increase ion tension. This increased tension applies organizational pressure to the chain — it narrows

the conformational windows available to each segment, biasing the chain toward compact, ordered configurations. Chaotropic ions do the opposite: they disrupt water structure, reduce ion tension, broaden the conformational windows, and allow the chain to sample extended, disordered configurations it cannot access under kosmotropic conditions.

This is not a direct chemical interaction between the ion and the protein backbone in most cases. Guanidinium does not covalently modify the protein. It destroys the water geometry that the folding landscape is built on. That is a fundamentally different mechanism than chemistry-first models typically invoke, and it explains why chaotropic denaturation is reversible — remove the chaotrope, restore the water geometry, and the folding landscape reasserts itself.

The cellular ion environment is not uniform, and this is not accidental. The cytoplasm is not a simple salt solution at fixed ionic strength. Local ion concentrations vary significantly based on proximity to ion channels, ion pumps, organelle membranes, charged cytoskeletal surfaces, and the local density of polyanionic RNA and DNA. Different proteins in different regions of the cell are operating under different local tension regimes. A protein folding near the ribosome exit tunnel is in a different ion environment than one folding in the cytoplasmic bulk, which is in a different environment than one folding near the inner leaflet of the plasma membrane. These are not equivalent folding contexts. The geometry-first view predicts that they should produce systematically different folding outcomes and folding speeds — and the existing data on compartment-specific protein behavior is consistent with this prediction, even if it has not been interpreted in these terms.

Ion tension also controls the speed of folding. A tighter, more organized water geometry constrains conformational sampling. It narrows the search space available to each segment. That sounds like a cost, but it is also a feature with clear functional logic: it prevents the chain from sampling catastrophically wrong conformations early in the folding trajectory, before the nucleation geometry has been established. Ion tension is a speed governor and a direction filter simultaneously. The cell controls it actively, through the coordinated activity of transporters, channels, and pumps, and the consequence is that the cell has

coarse-grained control over the folding landscape of the proteins it is synthesizing — not through direct chaperoning in every case, but through the medium those proteins fold in.

4. Fold Windows

A fold window is a narrow spatial and temporal zone in which a given segment of the polypeptide chain can commit to a stable configuration. The term is introduced here to name something that is physically real and mechanistically important but rarely discussed explicitly in folding theory. Outside the window — if the chain segment moves too fast, too slowly, in the wrong direction, or under the wrong tension state — it either fails to fold correctly or becomes trapped in a metastable non-native configuration. Fold windows are not arbitrary. They are defined by the intersection of three factors: the local water geometry at that segment's position, the ion tension state in the segment's immediate environment, and the existing geometry already established by segments that have already folded.

That third factor is critical and distinguishes folding from a simple equilibrium search. Folding is sequential. The chain emerges from the ribosome in a defined order, and even in post-translational folding, there are kinetically preferred nucleation regions that fold before others. Earlier-folding segments establish a geometric context that constrains later-folding segments. The fold window for residues 50 through 70 in a given protein is not determined solely by the intrinsic water geometry properties of those residues — it is determined jointly by those intrinsic properties and by the geometric boundary conditions imposed by whatever structure residues 1 through 49 have already established. This is a physically meaningful constraint. It means fold windows are context-dependent and history-dependent, which is a significant departure from equilibrium-based folding models that treat every point on the landscape as equally accessible from every other point.

Fold windows are not static. As the chain moves and conformational events occur, the water geometry around the chain changes continuously. Adjacent water shells are disturbed as segments collapse. Ion distributions shift locally as charged groups move

closer together or farther apart. The geometry of nearby segments responds to these changes. This means each folding event creates a new set of geometric conditions — a new set of fold windows — for subsequent folding events. The process is inherently dynamic and propagating, not an equilibrium search across a fixed landscape.

The Levinthal paradox becomes trivial in this framework. Cyrus Levinthal's 1968 observation — that a random search through all possible conformations of even a small protein would take longer than the age of the universe — has generated decades of theoretical work aimed at explaining why folding is fast. The canonical answer is that the energy landscape has a funnel shape, biased toward the native state, so the chain is not searching randomly. That answer is correct but incomplete. It describes the landscape without explaining what creates the bias or how the chain navigates it. The geometry-first answer is more specific: the chain is constrained at every step by the geometry of the water and ions around it. Fold windows eliminate the random walk entirely. At any given moment in the folding trajectory, the number of conformations the chain can physically access is not the full Levinthal number — it is the small subset of conformations compatible with the current geometric context. The chain is always moving in a biased field, and the bias is not an abstract thermodynamic funnel. It is a concrete, physically present geometric gradient in the water surrounding the chain.

The concept of fold windows also explains why folding is fast but not instantaneous. If the geometry perfectly determined the next folding event at every step, folding would be essentially barrierless. But the geometry is never perfectly constraining — there is always some conformational noise, some thermal fluctuation, some ambiguity in the gradient at any given position. The fold window has a width, not just a center. Crossing the window requires that thermal fluctuations align with the geometric gradient, which introduces a timescale — the timescale of folding — that is short compared to random search but long compared to the fastest molecular motions. That is exactly what is observed experimentally.

5. Propagation Mechanics

Folding propagates. It does not happen all at once across the entire chain. A nucleation event — the formation of the first stable secondary structure element — sets a geometric context that propagates outward through the chain. This propagation is not a metaphor borrowed from crystallization or phase transitions, though the analogy is instructive. It is a physically continuous process: the formation of a stable structural element changes the water geometry in a spatially continuous way through the surrounding medium, and that geometry change creates new fold windows for adjacent segments that did not previously exist.

The nucleation event in geometric terms works as follows. A short stretch of residues — typically four to seven amino acids — establishes a stable hydrogen-bonded configuration, most commonly the beginning of an alpha helix or the edge strand of a beta sheet. This event is significant not just because it establishes the first intramolecular contacts, but because it fundamentally changes the geometry of the water around those residues. Water is expelled from the interior of the nascent structure — there is simply no room for it inside a forming helix. That expelled water does not vanish. It reorganizes into a higher-order hydration shell around the exterior of the nascent structural element. This shell is mechanically distinct from bulk water. It is more ordered, under higher local tension, and it has a different spatial profile of pressure relative to the water that was there before nucleation.

This reorganized hydration shell around the nucleation site is geometrically continuous with the water surrounding adjacent chain segments. The chain is not in a vacuum — the water is a connected medium. A change in the geometry of water around residues 10 through 16 necessarily perturbs the geometry of water around residues 7 through 9 and 17 through 23. The perturbation is not uniform — it decays with distance and is modulated by the ion tension state — but it is real and physically significant on the length scales relevant to protein folding, which are nanometers, not microns. The adjacent segment now sits in a modified water geometry shaped by the existing nucleated structure. Its fold

window has moved. Conformations that were previously accessible are now geometrically disfavored because the surrounding water geometry no longer accommodates them. Conformations that were previously disfavored are now energetically accessible because the new water geometry creates gradients that bias the segment toward them. The chain samples within the new window and commits to the next stable element. This is propagation.

Propagation speed is controlled by ion tension. Under kosmotropic conditions, where ion tension is high, the geometric gradients established by nucleation events are steeper — the signal propagates more strongly into the adjacent water. The fold windows for subsequent segments are more precisely defined, and the chain commits to them faster. Under chaotropic conditions, where ion tension is low, the gradients are shallower, the windows are broader, and propagation is slower. This is consistent with experimental observations that kosmotropes accelerate folding of proteins that fold via nucleation-propagation mechanisms and that chaotropes slow or prevent it.

Propagation stalls are regions in the chain where the geometric gradient established by the existing structure is insufficient to clearly define the next fold window. These are kinetically slow regions — steps in the folding trajectory where the chain spends disproportionate time sampling conformations before committing. They are not random. They map onto sequence features that generate unusually permissive or conflicted water geometry. Proline residues are geometrically constrained in ways that resist both helical and extended beta conformations, making them natural stall points. Glycine residues, by contrast, are geometrically unconstrained — their water geometry is maximally permissive, providing little directional bias. Long loop regions, where the sequence encodes neither regular secondary structure nor strong hydrophobic burial, are typically weakly biased in their water geometry and are correspondingly slow to fold. Identifying propagation stalls from sequence should be possible in principle by calculating the geometric coherence of the water structure along the chain — the degree to which adjacent residues generate mutually compatible or conflicting water geometry requirements.

6. Wake-Flow Effects

When a segment of polypeptide chain folds — collapses, forms a helix, buries a hydrophobic patch, or undergoes any significant conformational change — it moves through the water. That movement is not passive. A folding chain segment is a macromolecular object displacing solvent at nanosecond timescales, and that displacement creates a local pressure wave in the surrounding water. Behind the moving segment — in the region from which it has just retreated — the water does not immediately return to its pre-disturbance equilibrium state. It passes through a transient reorganized configuration: a wake. The wake is not equilibrium water. It is a region of temporarily elevated or depressed structural organization whose specific properties depend on what kind of segment moved through it and in what direction it moved.

The timescale of wake decay is picoseconds to nanoseconds — water reorganization is fast by any macroscopic standard. But fast is relative. Conformational transitions in proteins that are relevant to folding — the movement of backbone dihedral angles, the formation of nascent secondary structure elements, the collapse of hydrophobic clusters — occur on timescales of nanoseconds to microseconds. This means a wake created by one folding event can persist long enough to influence the next folding event in an adjacent segment. The timescales overlap. Wake-flow effects are not negligible. They are part of the folding mechanism.

Wake-flow effects can be constructive or destructive, and the distinction matters for understanding both normal folding and misfolding. A constructive wake is one that improves the geometric context for a subsequent folding event. Consider a hydrophobic cluster collapsing away from the solvent — burying into the nascent hydrophobic core of a protein. As that cluster moves away from the surface, it leaves behind a wake of transiently organized water in the region it vacated. That organized water is rich in structured hydrogen bonds, relatively ordered, and presents a well-defined geometric context to the

adjacent polar or charged residues that now sit at the newly exposed surface. Those residues receive a constructive wake: the water geometry around them is temporarily enhanced, their fold window is more precisely defined, and they are more likely to commit correctly to their native configuration. The sequence of events — hydrophobic burial followed by polar surface exposure — is not just thermodynamically favorable; it is mechanically coordinated through wake-flow.

A destructive wake is one that degrades the geometric context for a subsequent folding event. A segment that moves in the wrong direction — toward the solvent when it should be burying, or into an incorrect conformation — creates a wake of disrupted water in the regions adjacent to its new position. That disruption flattens the geometric gradient for neighboring segments, broadening their fold windows and making incorrect conformations more accessible. Destructive wake-flow can propagate disruption downstream faster than the chain can recover, particularly under low-tension conditions where the water geometry is already weakly organized. This is one mechanism by which a single early misfolding event can cascade into a broader misfolding trajectory — not through direct structural contact with downstream residues, but through the mechanical perturbation of the water geometry in their vicinity.

The order of folding events therefore matters mechanically, not just thermodynamically. Two different sequences of folding events can, in principle, reach the same thermodynamic endpoint — the same native structure — but they differ in the wakes they generate along the way. The path with constructive wake-flow at each step is the path the chain preferentially takes, because the wake from each step improves the conditions for the next step. Natural folding sequences are ordered this way. The hierarchy of folding — secondary structure before tertiary packing, hydrophobic core collapse before surface loop placement — is not arbitrary. It is the order that produces the most constructive wake-flow sequence, minimizing destructive geometric perturbations and maximizing the clarity of fold windows at each successive step.

7. Misfolding as Geometry Drift

Misfolding is not random error. It is geometry drift — the progressive deviation of the water geometry around the chain from the geometry required for the native fold. This distinction is important. Random error implies that misfolding is an inevitable but unpredictable outcome of thermal noise and the complexity of the conformational search. Geometry drift implies that misfolding has specific physical causes, occurs through specific mechanisms, and is in principle predictable from knowledge of those mechanisms. The geometry-first view treats misfolding as a deterministic process, not a stochastic one, even though stochastic thermal fluctuations contribute to initiating it.

Geometry drift is initiated when a folding event — typically an early-stage event, before the chain has established a robust geometric framework — creates the wrong water geometry around a downstream segment. That wrong geometry biases the downstream segment into a non-native fold window. If the chain commits to that non-native window before the drift is corrected, the subsequent propagation proceeds from a deviated baseline. Each subsequent folding event follows geometric gradients that are correct relative to the deviated structure but incorrect relative to the native one. The chain folds coherently but into the wrong structure. This is geometry drift: the landscape has not become random; it has become organized around the wrong attractor.

The conditions that favor geometry drift are specific and predictable. High translation rates are one. When the ribosome elongates the chain faster than the water geometry can establish and equilibrate around each emerging segment, the cotranslational folding signal is lost — the chain emerges into a context where the geometry has not had time to set up the correct fold windows for the already-translated portion. This is one reason codon usage bias exists and matters. Synonymous codons that slow translation at specific positions give the water geometry time to establish, reducing drift. Incorrect ion tension is a second driver of geometry drift — if the local ion environment shifts during folding, the tension state

changes, the folding landscape shifts, and segments that were on track to native windows can be pulled into non-native ones by the changed gradients. Molecular crowding is a third driver: neighboring macromolecules perturb the water geometry around the folding chain in ways that are not encoded in its sequence, effectively adding geometric noise to the folding signal.

The distinction between kinetic traps and geometry drift is mechanistically important, because they require different corrections. A kinetic trap is a local free-energy minimum that slows folding — the chain is stuck in a configuration that is not the native state, but the surrounding geometry still favors the native fold if the chain is given sufficient thermal energy to escape the trap. The landscape has not changed; the chain is simply on the wrong side of a local barrier. Geometry drift is categorically different: it is a change in the surrounding geometry itself. The landscape has changed around the chain. The chain is following the geometry correctly — it is just following the wrong geometry. Kinetic traps are correctable by adding energy. Geometry drift requires resetting the geometry itself, which is a fundamentally different intervention.

This distinction is most relevant when considering amyloidogenic proteins. Amyloid-beta, tau, alpha-synuclein, TDP-43, and other proteins associated with neurodegenerative disease are not simply misfolded proteins in the sense of random structural errors. They are proteins whose sequences contain regions with water geometry properties that are weakly biased toward the native configuration and strongly biased toward flat, extended beta-sheet geometries under altered tension conditions. Under normal cellular ion tension, these regions fold into their native configurations — marginally stable, perhaps, but functional. Under altered tension conditions — oxidative stress, pH shifts, altered metal ion concentrations, aging-related changes in cytoplasmic ion composition — the geometric bias tips. The water geometry around these regions now favors beta-sheet configurations more strongly than native ones. The protein folds correctly into the geometry defined by the wrong tension state. This is not a failure to fold. It is folding working exactly as described, but working on a landscape that has been shifted by geometry drift into an

amyloidogenic basin. The disease state is not an accident. It is the geometry doing its job on the wrong input.

8. Aggregation Is Geometry Failure at Scale

Aggregation is what happens when geometry drift affects multiple chains simultaneously, and when the non-native geometry of one chain begins to impose its structure on the water surrounding neighboring chains. The critical insight here is that aggregation is not simply the consequence of misfolded proteins being sticky — having hydrophobic surfaces exposed and therefore adhesive. Stickiness is a chemical description of the endpoint. The mechanism by which misfolded chains recruit correctly folded or still-folding chains into the aggregate is geometric.

A misfolded chain presents a non-native surface geometry to the surrounding water. That surface geometry — for amyloidogenic proteins, typically an extended beta-sheet surface — restructures the local water in a characteristic way. The water adjacent to a flat, extended beta-sheet surface has a specific geometric organization: it forms a relatively ordered layer with particular hydrogen bonding geometry, tension state, and mechanical properties. This water layer is a template. It is physically present and geometrically organized, and it sits in the space through which neighboring chains must move to approach the misfolded aggregate. A soluble monomer diffusing toward an amyloid fibril surface encounters this geometrically organized water layer before it encounters any direct contact with the fibril. The organized water biases the monomer's conformation toward the beta-sheet configuration before the monomer touches the fibril. The fibril captures the monomer not by bonding to it but by reorganizing the water around it into a geometry that makes beta-sheet configuration the most favorable option. The fibril grows because the water does the geometry work before any protein-protein contact is established.

Concentration dependence follows directly from this mechanism. At low concentrations of misfolded chains, the non-native water geometry around any given misfolded chain is

overwhelmed by the bulk water around it — the geometric perturbation from a single misfolded chain is real but spatially limited, and soluble monomers that pass through the perturbed zone quickly re-enter bulk water and recover their native geometric context. At high concentrations, the non-native geometry fields of adjacent misfolded chains overlap spatially. The bulk water between misfolded chains is no longer geometrically neutral — it has been collectively shifted toward the non-native configuration by the contributions of multiple chains simultaneously. A monomer folding in this environment has no access to correctly organized bulk water. Its entire geometric context has been shifted toward the non-native configuration. It folds into the aggregate. The concentration threshold for aggregation is the threshold above which the non-native geometry fields of misfolded chains begin to overlap and collectively dominate the local water structure. This is a geometric phase transition, not a chemical one.

9. Chaperones as Geometry Reset Mechanisms

Molecular chaperones are proteins that assist other proteins in folding correctly, prevent aggregation, and rescue misfolded proteins. The canonical description is accurate. Hsp70, Hsp90, GroEL/GroES, and their relatives perform these functions in every living cell, and the experimental evidence for their activity is extensive and well-characterized. What is less well characterized is the mechanism by which they do it — specifically, the physical mechanism by which holding a misfolded protein and releasing it enables that protein to refold correctly. The geometry-first view provides a clear answer: chaperones are geometry reset mechanisms. They do not correct the sequence. They reset the water geometry around the client protein, restoring the geometric conditions under which the native fold windows are accessible.

The GroEL/GroES system is the best-characterized chaperonin and illustrates the geometry reset mechanism most clearly. GroEL is a large barrel-shaped complex that sequesters a misfolded client protein in a central cavity. The conventional description emphasizes that

the cavity provides a protected environment for refolding, preventing the client from interacting with other misfolded proteins and giving it an uninterrupted opportunity to reach the native state. This is true. But the geometry-first view adds the mechanistic detail: the interior of the GroEL cavity is not simply a generic protected space. It is a defined geometric environment. The cavity walls are lined with hydrophilic and charged residues arranged in a specific pattern. The water inside the cavity is geometrically distinct from bulk cytoplasmic water — it is more organized, under a different tension state, and spatially constrained by the geometry of the cavity walls. The client protein inside the cavity is not simply shielded from aggregation. It is immersed in a geometrically reset water environment that restores the tension state and geometric gradients required for native folding.

ATP hydrolysis and GroES binding change the geometry of the cage itself in a controlled, staged way. The GroEL cavity expands significantly when GroES binds. The charge distribution on the interior walls changes — residues that were previously pointing inward rotate or shift, presenting a different surface geometry to the enclosed water. This changes the ion tension inside the cage in a physically specific manner: the new wall geometry reorganizes the water inside the cavity and redistributes the ions within it, creating a new tension state. The result is a geometric program — a sequence of staged tension changes, driven by the ATPase cycle, that walks the client protein through the early folding windows again under conditions optimized to avoid drift. The ATP is not used to directly unfold and refold the client. It is used to execute the geometric program — to change the cavity geometry and therefore the water geometry in a specific sequence that mirrors the correct folding tension sequence.

Hsp70 operates on a smaller scale and with greater substrate specificity, but the geometric mechanism is analogous. Hsp70 binds to exposed hydrophobic segments — regions where hydrophobic residues are incorrectly solvent-exposed, creating the wrong water geometry for those residues and for adjacent segments. Hsp70 binding replaces the unfavorable water geometry around the exposed hydrophobic patch with a defined protein-protein interface.

This does two things geometrically: it removes the disrupted water geometry that the exposed hydrophobic surface was generating, and it physically shields that surface from the surrounding medium. When Hsp70 releases the client — triggered by nucleotide exchange from ADP to ATP — the previously bound segment re-emerges into the local water environment. That environment is no longer organized around an exposed hydrophobic surface. It is simply local cytoplasmic water at the current ion tension state, which is the correct starting geometry for the segment to attempt its fold window again. The geometry has been reset. The reset is local, fast, and reversible — appropriate for a chaperone that must service many clients rapidly under stress conditions.

10. Restoring Geometry: Implications and Predictions

If protein folding is fundamentally a geometry problem, then the practical implication is clear: restoring correct folding means restoring correct geometry. This is not merely a restatement of what chaperones do — it is a prediction about where effective interventions will be found, what they will look like, and how they will work. The geometry-first view generates testable, falsifiable predictions that are distinct from those of the chemistry-first view, and some of them bear directly on how diseases of misfolding should be treated.

The first prediction concerns pharmacological chaperones and small molecule interventions. Small molecules that stabilize the native conformation of a misfolded protein are already in clinical use — lumacaftor and tezacaftor for CFTR folding defects in cystic fibrosis being the clearest example. The chemistry-first interpretation of how these molecules work focuses on their binding to the protein and stabilizing the native structure through direct contacts. The geometry-first view predicts that molecules capable of tuning the local water geometry around specific misfolded segments — without necessarily binding deeply into the protein — should also be effective, and should in principle be designable from knowledge of the water geometry properties of the target segment. Osmolytes like trimethylamine N-oxide already work approximately this way — they do

not bind the target protein but restore its native fold by reorganizing the surrounding water. The prediction is that this class of mechanism is underexplored, and that molecules optimized for geometric water-tuning rather than direct protein binding represent an untapped therapeutic space.

The second prediction concerns the cellular ion environment as a folding control parameter. Changes in cytoplasmic ion composition — due to oxidative stress, metabolic dysfunction, aging, or disease — shift the tension state of the folding landscape broadly across many proteins simultaneously. The geometry-first view predicts that conditions causing widespread protein aggregation in neurodegeneration and other disease states should be accompanied by measurable changes in local ion concentrations and water structuring in the affected cell types, and that restoring ion homeostasis should have therapeutic effects beyond what can be explained by any single protein's chemistry. This is consistent with epidemiological observations about the protective effects of certain dietary ions and the role of metal ion dysregulation in Alzheimer's and Parkinson's disease, though the mechanistic interpretation has lagged behind the phenomenology.

The third prediction concerns cotranslational folding and synonymous codon usage. The geometry-first view treats cotranslational folding as a geometry synchronization problem: the ribosome must release the chain at a rate matched to the rate at which water geometry can establish around each emerging segment. Translation rate variation through codon usage bias is a mechanism for pacing this geometry establishment. This predicts that synonymous codon substitutions that alter translation speed at specific positions should change folding outcomes in ways that track with the water geometry requirements of the affected segments — not uniformly, but specifically at positions where the geometry establishment timescale is rate-limiting. This is a concrete, testable prediction that goes beyond the current framing of codon bias effects on folding, which remains mechanistically vague.

The fourth prediction concerns protein evolution. The geometry-first view holds that evolution has been selecting not just for thermodynamic stability of the native fold but for

geometric coherence along the chain — the degree to which adjacent residues generate mutually compatible, reinforcing water geometry requirements that produce clean, well-directed fold windows. Sequences with high geometric coherence fold fast, fold reliably, and resist aggregation under stress. Sequences with geometric incoherence — where adjacent residues generate conflicting or weakly directed water geometry — fold slowly, misfold frequently, and aggregate readily when cellular conditions change. This predicts that evolutionary conservation of sequence positions should correlate with their contribution to geometric coherence, not just with thermodynamic contributions to stability. Residues that are not in the hydrophobic core and do not make obvious structural contacts but are nevertheless highly conserved may be conserved for geometric coherence reasons — they set the water geometry context for adjacent segments in ways that are invisible to structure-based analysis but critical for the folding mechanism.

11. Conclusion

Protein folding is a geometry problem. The sequence determines structure because the sequence determines the water geometry around the chain, and the water geometry defines the landscape the chain folds across. Ion tension sets the rules governing how tight that landscape is, how fast folding propagates, and how far fold windows are from each other. Fold windows are real, physical, and calculable in principle from the geometry of the water surrounding each segment. Propagation is mechanical — it moves through the water as a continuous medium, driven by the geometric gradients established by each successive folding event. Wake-flow effects are real, order-dependent, and mechanistically significant on timescales relevant to folding. Misfolding is geometry drift. Aggregation is geometry drift at scale, driven by the capacity of misfolded surfaces to reorganize surrounding water into non-native templates. Chaperones reset geometry — they restore the water geometry and ion tension context in which native fold windows are accessible, either through enclosed cavity environments or through targeted shielding of incorrectly exposed surfaces.

None of this requires discarding the chemistry-first view. Hydrogen bonds, hydrophobic contacts, and electrostatic interactions are all real forces, and they are all present in any complete account of protein folding. The argument here is that these forces are downstream outputs of a geometric program running in the water and ion field surrounding the chain. Chemistry provides the specificity of individual contacts. Geometry provides the landscape on which those contacts are made. The chemistry-first view of folding has produced enormous progress — sequence databases, energy functions, and structure prediction tools that work with remarkable accuracy. AlphaFold2 predicts native structure from sequence with a precision that is genuinely stunning. But AlphaFold2 is an input-output machine. It does not model mechanism. It does not explain why a sequence folds to a given structure, only that it does. Knowing that a sequence folds to a given structure does not tell you how to prevent that structure from drifting under cellular stress, how to rescue the protein when it does drift, what the cell is doing to maintain thousands of correctly folded proteins under variable and challenging conditions, or why specific sequences are inherently amyloidogenic while others of similar thermodynamic stability are not. Geometry does.

The view developed here is testable. The predictions are specific. The mechanism is physical and continuous with established physics of water and ions. The next step is to measure it directly — to map the water geometry around folding chains in real time, to perturb ion tension in controlled and spatially localized ways, and to correlate those measurements with folding trajectories at single-molecule resolution. The experimental tools to begin doing this exist. The conceptual framework is here. Start there.